Case Study: Song Filtering of YouTube Music

Title:

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Overview:

This case study focuses on a system designed to filter and play songs efficiently on YouTube Music. With massive song libraries, users often struggle to locate specific tracks quickly. The project provides a simple solution for filtering songs by categories or names and enabling smooth playback.

Introduction:

Music streaming platforms like YouTube Music have extensive song collections, making navigation challenging. While search and playlists exist, filtering songs by categories such as genre, artist, or mood enhances user experience. The "Song Filter" project addresses this need.

Problem Statement:

Users face difficulty in quickly finding desired songs in large collections. Existing tools may be insufficient for filtering based on multiple criteria, causing wasted time and frustration. A more precise and interactive filtering system is required.

Methodology:

Requirement Analysis: Studied YouTube Music's interface and filtering options.

Design: Created a system to filter songs by name, genre, or category.

Implementation: Developed the program for filtering and playback using sample song data.

Testing: Verified filtering efficiency and playback functionality with multiple test cases

Proposed Solution:

The “Song Filter” system offers:

- Filtering songs by genre, artist, or mood.
- Search functionality to locate specific songs by name.
- Smooth playback of filtered songs.
- A simple, interactive interface for user convenience.

Discussion:

Filtering reduces search time and increases efficiency.

Playback works seamlessly with filtered results.

Limitations include dependence on data quality and handling large datasets.

Future improvements could include dynamic filtering options, integration with live YouTube data, and better scalability.

Presentation:

Include:

- Screenshots of program interface or GitHub repository.
- Sample filtered song lists and playback outputs.
- Workflow diagram if available.

Conclusion:

The project successfully addresses the challenge of locating and playing songs efficiently on YouTube Music. Filtering enhances user experience and makes navigation faster and intuitive. Future enhancements can expand functionality and improve scalability.

GitHub Repository:

GitHub link: https://github.com/sayli-borade/song-filter-playback